Reviewer 2 Report — Critical Analysis of "Training-Free LLM-Generated Text Detection by Mining Token Probability Sequences" (ICLR 2025)

Reviewer posture: strict ICLR Reviewer 2. All claims below are grounded in the paper's own text, figures, and tables (cited as Section/Figure/Table); anything the paper does not report is explicitly marked "Not reported in the paper." Where I derive quantities from the algorithm's description (e.g., computational complexity), this is flagged as *derived analysis*, not as a paper-reported number.

Overall recommendation: Weak accept, contingent on a major-revision cycle. The paper's core contribution — introducing time-series analysis of the token probability sequence — is original and its empirical gains over the training-free family are consistent and large (Tables 1–2). However, four concerns materially limit the claims the paper can make: (1) the black-box detector is hostage to an arbitrary proxy choice and the statistic is semantically blind; (2) the "efficiency" narrative is never measured; (3) the adversarial evaluation does not cover the two most realistic threat models (human editing, mixed authorship); (4) the central empirical law ("humans fluctuate more") has no mechanistic foundation and is plausibly invertible by decoding control.

Concern 1 (Methodological): Proxy-Model Dependence Is the Method's Achilles' Heel, and DE Is Blind to Global Semantics

Analysis. In the black-box setting, every score is computed against a *fixed* proxy model — GPT-J for English/German and Yi1.5-6 for Chinese (Section 4.2) — so $\mathrm{Lastde}(t,\theta)$ (Eq. 9) and the standardized discrepancy $D(t,\theta,\tilde{\theta})$ (Eq. 10) are functions of an arbitrarily chosen model. The paper reports *what* proxy was used, but provides no principled criterion for *choosing* one; the choice is treated as given (Section 4.2). That this is not a cosmetic issue is shown by the paper's own Figure 8: detection performance varies strongly across (source, proxy) pairings, and in the hardest pairing (source GPT-4-Turbo, proxy Llama3-8) *all* methods degrade substantially. The Ethics Statement concedes the point directly: detection "still produce[s] imperfect detection results when faced with sophisticated challenges such as paraphrasing attacks and the arbitrary replacement of proxy models."

Why this is severe. Black-box is the realistic deployment regime — it is precisely where proprietary models (GPT-4o, Claude) are used, and where the paper's headline advantage (+5.80 AUROC over Fast-DetectGPT, Table 2) is largest. But the adversary has perfect knowledge of the fixed public proxy: they can generate with a source model far from it, or tune their generator to maximize disagreement with the proxy. The detector's strongest result is therefore hostage to a variable the paper neither controls nor optimizes, and the paper offers no error bound or selection procedure. The second, subtler defect compounds the first: the DE pipeline (Eqs. 2–8) consumes *only* the log-probability values $-\log p_\theta(t_i \mid t_{<i})$. Token identities, sentence structure, and discourse-level coherence never enter the computation. With the default window $s=3$ and scales up to $\tau'=5$ (Section 3.2), the "local view" spans a handful of tokens, so long-range semantic structure — argument flow, topic drift, factual coherence — is invisible to the detector. DE measures the *geometry of the probability trace*, not the *meaning of the text*: a semantically incoherent but model-confident text scores "smooth" (LLM-like), while a coherent human text with volatile word choices scores "fluctuating" (human-like). The statistic and the construct it is meant to detect (authorship) are connected only empirically.

Potential solution. (a) Replace the fixed proxy with *proxy selection or aggregation*: choose θ by minimizing perplexity or calibration error on an unlabeled sample of the deployment domain, or ensemble several proxies and aggregate the Lastde/Lastde++ scores (a cheap extension, since each score is already a single pass). (b) Augment the local statistic with *semantic features*: sentence-embedding coherence, discourse-level hierarchical TPS analysis (DE computed on sentence- or paragraph-aggregated probability traces), which restores access to global structure while remaining training-free.

Concern 2 (Computational Cost): "Training-Free" Is Marketed as "Efficient," but Inference Latency Is Never Measured

Analysis. The paper reports no wall-clock latency, throughput, or FLOP accounting anywhere: the actual inference latency of Lastde++ relative to the simplest Log-Likelihood baseline is **not reported in the paper**. The only hardware disclosure is the experimental setup "consist[ing] of two RTX 3090 GPUs (2×24GB)" (Appendix B.2). Every efficiency claim is *indirect*: 100 samples versus Fast-DetectGPT's 10,000 (Section 3.2); Figure 4, showing 10 Lastde++ samples surpassing Fast-DetectGPT's 100; and the observation that unifying sampling and scoring into one model "saves GPU memory" (Appendix B.2) — a memory remark, not a time measurement. Meanwhile, the MDE computation itself is the most involved statistic in the field. *Derived analysis* from the algorithm description in Section 3.1: for each scale τ, the pipeline computes $n\text{-}s+1$ sliding-window cosine similarities over s-dimensional segments — $O(n\cdot s)$ — plus a histogram over $\varepsilon=10n$ (Lastde) or $\varepsilon=8n$ (Lastde++) bins — $O(n)$; across τ' scales the total is $O(\tau'\cdot n\cdot s)$ versus $O(n)$ for plain Log-Likelihood. With the defaults $s=3,\tau'=5$ (Lastde) and $s=4,\tau'=15$ (Lastde++), that is roughly **15× and 60× the per-token arithmetic** of the likelihood baseline, *before* adding 100 autoregressive sampling passes, whose sampled continuation length is not specified (the fast-sampling scheme is borrowed from Bao et al., 2024, and not re-derived, Section 3.2).

Why this is severe. The paper's abstract-level selling point is that Lastde++ needs only $1/100$ of Fast-DetectGPT's samples — a claim that reads as "near-real-time." But 100 generations is still 100 full model calls, two orders of magnitude more than a single-score detector, and the newly introduced statistic multiplies per-token cost by an order of magnitude relative to Likelihood. For real-time use (e.g., per-message screening, API-based scoring at scale), no practitioner can decide feasibility from this paper, because the one quantity that matters — absolute latency — is absent. "Training-free" removes the *offline* cost; the paper never establishes the *online* cost.

Potential solution. (a) Reduce constant factors without losing signal: strided windows (sample every k-th segment instead of stride 1), a fixed-size histogram (constant ε, e.g., 128 bins, rather than $\varepsilon\propto n$), and dropping large τ scales for short texts — consistent with the paper's own finding that short texts restrict usable scales (Figure 5). (b) Cut sampling from 100 to the ~10 suggested by Figure 4. (c) Report a standard latency/throughput benchmark of Likelihood, Lastde, Lastde++, and Fast-DetectGPT on identical hardware; this single addition would turn the efficiency narrative from rhetorical to falsifiable.

Concern 3 (Real-World Robustness): Human Editing and Mixed Human–AI Authorship Are Untested

Analysis. The robustness suite covers three adversarially-motivated cases: machine paraphrasing via the T5-Paraphraser (Table 3), "human-style mimicking" via word-level editing toward human-like phrasing (Appendix F.4, Table 16), and *LLM–LLM* mixture texts built by truncation-concatenation of 2 and 4 LLMs (Appendix F.5, Tables 17–19). What is missing matters more than what is present. (i) **Heavy human paraphrasing** — a person substantially rewriting, restructuring, or translating AI text — is **not reported in the paper**; T5-paraphrasing is a weak proxy for human revision, and the paper's own Table 3 shows even that weak attack costs Lastde++ up to −9.14 AUROC (black-box XSum). (ii) **Mixed human–AI co-authored text** — the dominant real-world pattern (AI drafts, human edits; interleaved human/AI paragraphs) — is **not reported in the paper**; the mixture experiments (Tables 17–19) concern LLM–LLM concatenation only, and the paper offers no evidence on detecting *partial* AI contribution within one document. (iii) **Short texts**: Figure 5 shows degradation below ~120 words because fewer smoothing scales τ' remain usable, and the authors excluded PubMedQA precisely because its samples are "significantly shorter" (Appendix B.1) — yet short texts (social-media posts, chat messages) are the highest-stakes detection setting. (iv) **Thresholding**: Appendix G fits practical thresholds with logistic regression on labels — which the authors themselves admit "appears to undermine the 'training-free' aspect" — and even with fitted thresholds, TPR@1%FPR is only 52.33% (vs. 32.62% for Fast-DetectGPT), meaning roughly half of AI text escapes at a strict false-positive budget.

Why this is severe. Real-world deployment differs from this evaluation along exactly the untested axes. The paper's robustness victories are *relative* ("loses the least," Table 3), not absolute, and the one case that defines the policy question — did AI contribute to this human-finished document? — has no evidence at all. A detector evaluated only on machine paraphrase and machine mixtures tells us little about academic-integrity or content-moderation settings, where human revision and partial AI use are the norm.

Potential solution. (a) Segment-level scoring: compute Lastde on sentence/paragraph units and aggregate (e.g., max or fraction above threshold) to detect partial AI contribution. (b) Build and evaluate a *human-edited-AI* corpus (e.g., crowdworkers rewriting AI drafts) and interleaved human/AI documents. (c) Extend the label-free pseudo-label threshold scheme sketched around the ≈2.0 score boundary (Appendix G) and validate it out-of-domain, removing the label dependence the authors themselves flag.

Concern 4 (Theoretical Gap): "Humans Fluctuate More" Is an Observation, Not a Law — and Decoding Control Can Plausibly Invert It

Analysis. The paper's entire method rests on one empirical regularity: human TPS curves lie above LLM curves at every smoothing scale (Figure 9). The only justification offered is that LLM-generated continuations "are more compact, whereas human responses exhibit the opposite trend," attributed to LLMs being "trained with the objective of minimizing perplexity" (Appendix E). This is an *intuition about training objectives*, not a theorem: the paper provides no formal conditions under which $\mathrm{DE}_{\mathrm{human}}(s,\varepsilon,\tau) > \mathrm{DE}_{\mathrm{LLM}}(s,\varepsilon,\tau)$, and no analysis of how DE responds to the model's decoding parameters. The gap is established on the authors' corpora under their generation protocol (150 + 150 samples per dataset, 30-token prefix, Appendix B.1) — nothing in the paper bounds its persistence under distribution shift. The decoding-robustness experiment (Figure 6, Table 15 in Appendix F.3) tests ordinary settings — top-p = 0.96, top-k = 40, temperature = 0.8 — and shows per-dataset averages above 94%, but it does *not* probe the regime that decides the theoretical question:

how DE varies with temperature at all, or at extreme/adversarial settings, is **not reported in the paper**. This is a real gap, because DE confounds author identity with model confidence: greedy decoding ($T\to 0$) flattens the TPS (low DE, "LLM-like" by construction), while high-temperature sampling or noise-token injection inflates fluctuation (high DE, "human-like") — the sign of the discriminative gap is under the generator's control. Relatedly, the score is a deterministic, cheap-to-evaluate function of the text through a public proxy model, so it is in principle amenable to *adaptive* optimization-based attacks (iterative, search- or gradient-guided rewriting that minimizes $\mathrm{Lastde}(t,\theta)$ or $D(t,\theta,\tilde{\theta})$); the paper evaluates only static, off-the-shelf paraphrasers (Table 3). Adaptive attacks are **not reported in the paper**.

Why this is severe. A detector whose decision rule rests on an unexplained empirical regularity cannot predict its own failure modes. Without a mechanistic account of *why* DE separates the classes, we cannot know whether the separation survives (a) adversarial decoding parameters, (b) models fine-tuned to mimic human writing statistics, or (c) an adaptive adversary who treats the proxy as a loss function. The paper's robustness results, being limited to static attacks and standard decoding, leave precisely these scenarios open — and each of them is cheap for a motivated adversary to execute.

Potential solution. (a) Formal analysis linking DE to per-token predictive entropy and sampling temperature (e.g., bounding $\mathbb{E}[DE]$ under greedy versus high-temperature sampling, and its monotonicity in T), which would either certify the current feature or expose the inversion regime. (b) Adversarial evaluation: REINFORCE-style or search-based rewriting against the proxy, and detection of texts generated with deliberately extreme decoding parameters. (c) Defensively, supplement DE with signals whose temperature-dependence is understood — calibrated perplexity, rank-based statistics (LogRank), or combinations — so that the detector does not depend on a single unproven regularity.

Summary Table

#	Concern	Core evidence (paper)	Severity	Suggested direction for Task 5
1	Proxy dependence + semantic blindness	Figure 8; Ethics Statement; Eqs. 2–8 use only $\log p_\theta$; $s=3,\tau'=5$ (Section 3.1–3.2)	Fatal in black-box; undermines the setting where the method wins most	Proxy selection/ensembling; semantic-aware (discourse/embedding) features
2	Unmeasured inference cost	No latency reported; 2×RTX 3090 (Appendix B.2); derived $O(\tau'\cdot n\cdot s)$ vs $O(n)$; 100 sampling passes	Blocks any real-time deployment claim	Constant-factor reductions (strided windows, fixed bins); latency benchmarks; ~10-sample mode per Figure 4
3	Untested real-world robustness	Only T5 paraphrase (Table 3), word-level mimicking (Table 16), LLM–LLM mixtures (Tables 17–19); Figure 5 short-text weakness; label-fitted thresholds (Appendix	Human editing & mixed authorship are the dominant real threat models; no evidence	Segment-level scoring; human-edited-AI corpora; label-free thresholds

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4	No mechanistic theory; temperature/adaptive-attack exposure	"Minimizing perplexity" intuition (Appendix E); decoding tests only at standard settings (Table 15, Appendix F.3); no adaptive attacks	Decision rule rests on an unproven regularity that the generator may control	Temperature–DE formal analysis; adaptive-attack evaluation; hybrid statistics

Bottom line for the authors: the contribution is real and the empirical results are strong within the training-free family, but the paper overclaims relative to its evidence on exactly four fronts — proxy robustness, efficiency, real-world adversarial coverage, and theory. All four are addressable with targeted experiments and modest methodological extensions, several of which the paper already gestures at (label-free thresholds in Appendix G, the 10-sample regime in Figure 4). These four gaps, and their candidate fixes, are the natural starting points for designing our team's next approach (Task 5).